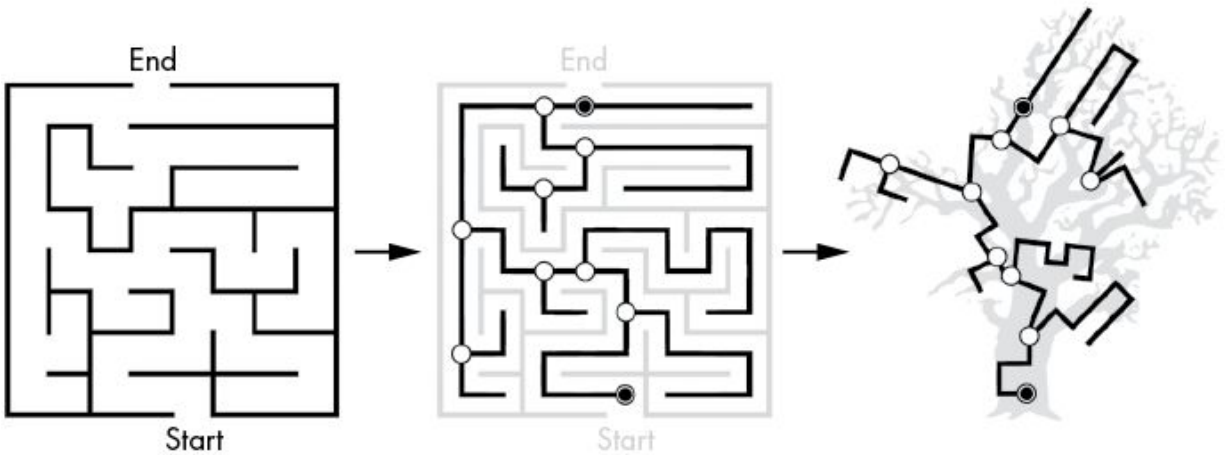
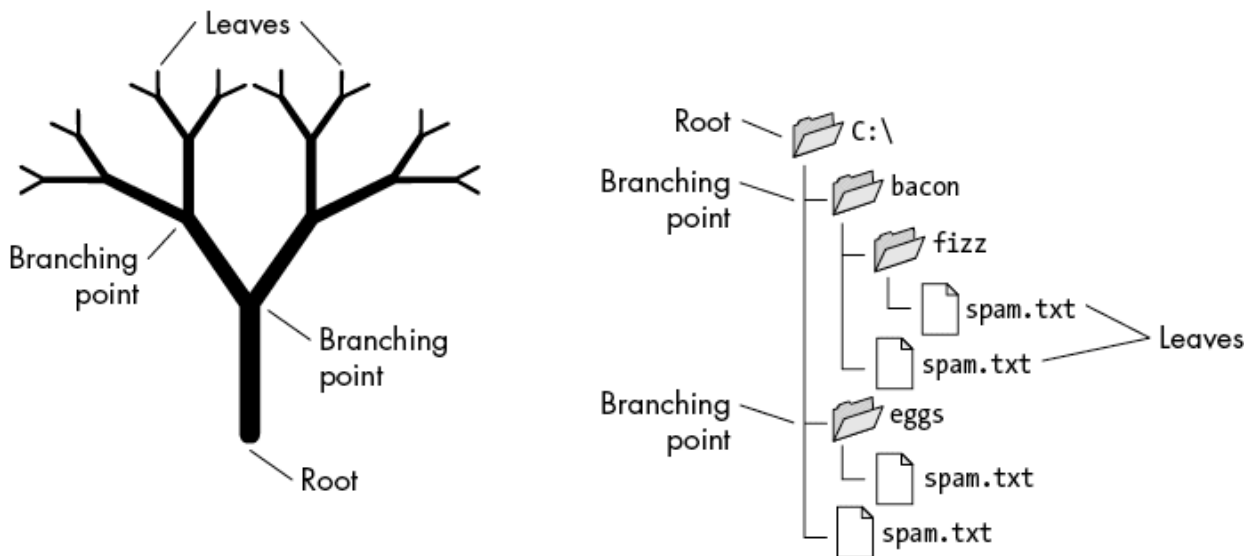




*Figure 2-5: A maze (left) along with its interior paths (center) morphed to match a biological tree's shape (right)*

Many programming problems have this tree-like structure at their core. For example, a filesystem has a tree-like structure; the subfolders look like the root folders of a smaller filesystem. [Figure 2-6](#) compares a filesystem to a tree.



*Figure 2-6: A filesystem is similar to a tree structure.*

Searching for a specific filename in a folder is a recursive problem: you search the folder and then recursively search the folder's subfolders. Folders with no subfolders are the base cases that cause the recursive searching to stop. If your recursive algorithm doesn't find the filename it's looking for, it backtracks to a previous parent folder and continues searching from there.